

Maximus Fernandez

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Education

University of South Carolina, Columbia, SC

Expected May 2027

B.S. in Computer Science, Minor in Business Administration

GPA: 3.5

Coursework: Software Engineering, Database System Design, Data Structures & Algorithms, Adv. Programming in C++

Projects

SC Economics Volunteer Management System, Kappa Theta Pi @ USC

January 2026 – April 2026

- Built RESTful API routes with Prisma and Neon Postgres, enforcing role-based access control across 3 user tiers via a 2-layer authentication and authorization proxy.
- Designed and implemented backend logic for application review workflows, event CRUD with signup/withdrawal transactions, and volunteer hours logging, all gated behind server-side staff session validation.
- Developed idempotent seed/unseed scripts with shared-database safety guarantees and deployed the application to Vercel with environment-managed database connections, enabling safe team demos without risk to production data.

InsureFlow AI, Palmetto Hacks

April 2026

- Engineered FNOL orchestration and triage APIs using FastAPI, Python, and Next.js API routes that forwarded damage photos to Azure AI Vision for automated severity classification, cutting simulated claim intake from hours to minutes.
- Integrated GPT-4o with structured JSON prompts for underwriting field extraction from broker PDFs and copilot-assisted form completion, persisting claims and adjuster review decisions in PostgreSQL with session-based authentication.
- Collaborated across a 4-person team to build human-in-the-loop approve, override, and escalate workflows with explainable confidence scoring and audit logging, earning 2nd place in the Capgemini insurance operations challenge.

SmartRoute, GridStorm Hackathon

March 2026

- Built a 4-stage document triage pipeline for Dominion Energy using OpenCV and Tesseract for OCR preprocessing, Mistral (Ollama) for LLM field extraction, Pydantic v2 for schema validation with confidence scoring, and a Python rules engine routing records to safety, field ops, scheduling, or human review.
- Implemented a Streamlit-based operator dashboard with confidence-weighted scoring and human review overrides, enabling real-time correction and re-routing of flagged records.

Leadership

Technical Events Coordinator, Association for Computing Machinery (ACM)

September 2024 – Present

- Organized 10+ technical workshops and speaker sessions, scaling attendance to 20+ per session.
- Mentored 20+ peers through LeetCode and DSA interview preparation and designing reusable workshop materials.

Experience

IT Intern, Validation Services Team, BlueCross BlueShield, Columbia, SC

May 2025 – August 2025

- Built automated browser regression checks using Selenium (XPath), validating navigation flows, search result counts, and form inputs across multiple site which reduced manual testing time by 60%.
- Executed 30+ manual test cases across web, mobile, and IVR systems during a Genesys Cloud migration, validating call routing logic and edge-case handling as part of the team's SDLC process.
- Tracked defects and coordinated test progress through Azure DevOps pipelines, incorporating QA lead feedback on test coverage gaps to refine automation scripts before release.

Undergraduate Research Assistant, SyReX @ USC, Columbia, SC

January 2024 – June 2024

- Built automated preprocessing pipelines for multimodal sensor datasets, reducing manual work by 45% for downstream Fourier-based signal analysis.
- Developed MATLAB and Python tooling to capture and synchronize audio/video data from human-subject experiments, supporting non-invasive heart and respiration monitoring research using FMCW acoustic signals.

Skills

- **Languages:** Java, Python, TypeScript, JavaScript, C++, SQL, Bash, MATLAB, HTML, CSS
- **Frameworks & Libraries:** Next.js, FastAPI, Prisma, Tailwind CSS, React, Streamlit
- **Tools & Platforms:** Git, GitHub, Selenium, Azure, PostgreSQL, MySQL
- **Core Concepts:** RESTful APIs, Object Oriented Programming (OOP), SDLC, Agile, Scrum